

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)

Running a Standard Simulation with GEOS-Chem

Learning Objectives:

- Understand the fundamental workflow of GEOS-Chem Classic, a global 3D chemical transport model used to simulate atmospheric composition.
- Use the `createRunDir.sh` script to configure a short-period, standard global full-chemistry simulation at a coarse resolution.
- Locate and interpret the primary NetCDF output files and ensure the simulation completed successfully with physically plausible results.

CUHK Central Research Computing Cluster

Instructions: <https://www.cuhk.edu.hk/itsc/hpc/getstart.html>

Running GEOS-Chem on CUHK Central Research Computing Cluster

The following workflow is adapted from TGABI Group's manual, developed with the effort of Lina Luo, Yanzhi Wang, Prof. Shixian Zhai, and Zheng Yu, and serves as a quick-start guide for users to run their first GEOS-Chem simulation.

1. Jump to Sandbox

```
ssh sandbox
```

2. Create Working Directory in EESC4550's Project Folder

```
cd /lustre/project/eesc4550/  
mkdir <COMPUTING_ID>
```

```
[s1155189917@chpc-logina ~]$ ssh sandbox  
The authenticity of host 'sandbox (10.68.9.104)' can't be established.  
ECDSA key fingerprint is SHA256:aWfASKKJ3SNhCURUwaoLVwg+geT9h/JQ5hdFbfNuedpY.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'sandbox,10.68.9.104' (ECDSA) to the list of known hosts.  
[s1155189917@chpc-sandbox ~]$ cd /lustre/project/eesc4550/  
[s1155189917@chpc-sandbox eesc4550]$ mkdir s1155189917
```

3. Downloading Source Code, Compiling and Test Running

Download GEOS-Chem to your personal directory with the following command:

```
cd /lustre/project/eesc4550/<COMPUTING_ID>  
mkdir GEOS-Chem  
cd GEOS-Chem  
git clone --recurse-submodules https://github.com/geoschem/GCClassic.git  
my_code_dir  
Receiving objects: 100% (711/711), 698.32 KiB | 1.04 MiB/s, done.  
Resolving deltas: 100% (462/462), done.  
Submodule path 'src/HEMCO/docs/source/geos-chem-shared-docs': checked out '3887938a14f933607b597e04436e1741f102d017'  
Submodule path 'src/HETP': checked out '2a99b24625ed26cf87ae88697ddd6cf8bbdec812'  
[s1155189917@chpc-sandbox GEOS-Chem]$ cd my_code_dir/  
[s1155189917@chpc-sandbox my_code_dir]$ ls  
AUTHORS.txt  CHANGELOG.md  CMakeLists.txt  CMakeScripts  CONTRIBUTING.md  docs  LICENSE.txt  README.md  run  spack  src  SUPPORT.md  test  
[s1155189917@chpc-sandbox my_code_dir]$
```

4. Create a Run Directory

```
mkdir GC_rundirs  
cd my_code_dir  
cd run  
./createRunDir.sh
```

You will then be prompted for the following options:

- ExtData path: `/project/eesc4550/GEOS-Chem/ExtData` (This is a shared folder for this course and stores the test case data; please do not change anything in this folder)
- Initiating User Registration: Please type in the answer required by registration.
 - What is your name (first and last)?
 - What is your email address?
 - What is the name of your research institution?
 - What is the name of your principal investigator
 - Please provide the web site for your institution
 - Please provide your github username (if any) so that we can recognize you in submitted issues and pull requests.
 - Where do you plan to run GEOS-Chem? (e.g. local compute cluster, AWS, other supercomputer)?
 - Please briefly describe how you plan on using GEOS-Chem so that we can add you to 'GEOS-Chem People and Projects' (<https://geoschem.github.io/geos-chem-people-projects-map/>)
- Simulation Type: `1` (Full chemistry)
- Additional Simulation Option: `2` (Benchmark)
- Meteorology Source: `1` (MERRA-2)
- Horizontal resolution: `1` (4.0×5.0)
- Number of levels: `2` (47 (reduced))

Path where the run directory will be created: `/project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs`

- Run directory name: Set a name or press 'enter' to use the default name
- Track changes with git: `y` (Please do so since we will use git for future code revisions)
- Do you want to build the KPP-Standalone Box Model? (y/n): `y`

```
>>>> REMINDER: You must compile with options: -DKPPSA=y <<<<
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <s1155189917@chpc-sandbox.rc.cuhk.edu.hk>) not allowed

Created /ss02/data/project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark
[s1155189917@chpc-sandbox run]$
```

Remarks: Note that these options only affect the run directory contents, and NOT the build process - the same GC-Classic executable is usable for almost all simulation types and supported met data options.

5. Copy the Slurm Script and an Environment File to the Run Directory:

```
cd /project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs
ls
cd gc_4x5_47L_merra2_fullchem_benchmark
cp -f /project/eesc4550/GEOS-Chem/scripts/* .
```

For global run:

```
ln -s
/lustre/project/eesc4550/shared/global/GEOSChem.Restart.20190701_0000z.nc4 ./Restarts/GEOSChem.Restart.20190701_0000z.nc4
```

```
[s1155189917@chpc-sandbox run]$ cd /project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs
[s1155189917@chpc-sandbox GC_rundirs]$ cd gc_4x5_47L_merra2_fullchem_benchmark/
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ cp -f /project/eesc4550/GEOS-Chem/scripts/*
cp: target '/project/eesc4550/GEOS-Chem/scripts/run.slurm' is not a directory
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ cp -f /project/eesc4550/GEOS-Chem/scripts/* .
cp: overwrite './HISTORY.rc'? y
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ ls
archiveRun.sh  CodeDir  download_data.py  geoschem.benchmark.run  HEMCO_Config.rc  HISTORY.rc  kpp_standalone_interface.yml  OutputDir  runScriptSamples
build  compile.slurm  download_data.yml  geoschem_config.yml  HEMCO_Config.rc.gnaw_metfields  README.md  run.slurm  species_database.yml
cleanRunDir.sh  CreateRunDirLogs  gcc.env  getRunInfo  HEMCO_Diagn.rc  metrics.py  Restarts
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$
```

For nested-grid run:

```
ln -s
/lustre/project/eesc4550/shared/nested/GEOSChem.Restart.20190701_0000z.nc4 ./Restarts/GEOSChem.Restart.20190701_0000z.nc4
```

6. Compiling GEOS-Chem

Run the following command on the sandbox:

```
bash compile.slurm |& tee compile.log
```

```
[ 1%] Built target KPP_FirstPass
[ 4%] Built target CloudJ_core
[ 13%] Built target Headers
[ 13%] Built target JulDay
[ 18%] Built target NcdFutil
[ 24%] Built target GeosUtil
[ 25%] Built target ObsPack
[ 27%] Built target HeadersHco
[ 28%] Built target JulDayHco
[ 33%] Built target NcdFutilHco
[ 33%] Built target GeosUtilHco
[ 46%] Built target HCO
[ 50%] Built target HCOX
[ 57%] Built target HCOI_Shared
[ 59%] Built target HETP_core
[ 68%] Built target KPP
[ 71%] Built target History
[ 98%] Built target GeosCore
[100%] Built target gcclassic
Install the project...
-- Install configuration: "Release"
-- Installing: /project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark/build_info/CMakeCache.txt
-- Installing: /project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark/build_info/summarize_build
-- Installing: /project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark/gcclassic
-- Set runtime path of "/project/eesc4550/s1155189917/GEOS-Chem/GC_rundirs/gc_4x5_47L_merra2_fullchem_benchmark/gcclassic" to ""
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$
```

Revise the setting: `start_date`, `end_date`

```
vi geoschem_config.yml
```

```
---
### geoschem_config.yml: GEOS-Chem Runtime configuration options.
### Customized for simulations using the KPP "fullchem" mechanism.
###
### NOTE: Add quotes around nitrogen oxide ('NO'), because YAML
### parsers will confuse this with a negative "no" value.

#=====
# Simulation settings
#=====
simulation:
  name: fullchem
  start_date: [20190701, 000000]
  end_date: [20190703, 000000]
  root_data_dir: /project/eesc4550/GEOS-Chem/ExtData
  met_field: MERRA2
  species_database_file: ./species_database.yml
  species_metadata_output_file: OutputDir/geoschem_species_metadata.yml
  verbose:
    activate: false
    on_cores: root # Allowed values: root all
  use_gcclassic_timers: true
```

7. Submitting a Job to GEOS-Chem

Submit the job to the slurm system:

```
sbatch run.slurm
```

```
ls
```

```
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ sbatch run.slurm
Submitted batch job 1827010
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ ls
1827010_error.log  build_info  compile.slurm  gcc.env  getRunInfo  HISTORY.rc  README.md  species_database.yml
archiveRun.sh     cleanRunDir.sh  CreateRunDirLogs  gccClassic  HEMCO_Config.rc  kpp_standalone_interface.yml  Restarts
build             CodeDir        download_data.py  geoschem.benchmark.run  HEMCO_Config.rc.gnss_netfields  metrics.py  runScriptsSamples
build            compile.log    download_data.yml  geoschem_config.yml  HEMCO_OIagn.rc  OutputDir  run.slurm
```

```
tail -f <job-id>_print_out.log
```

```
- Found all A1      met fields for 2019/07/01 10:30
--> DATE: 2019/07/01 UTC: 10:10
--> DATE: 2019/07/01 UTC: 10:20
--> DATE: 2019/07/01 UTC: 10:30
--> DATE: 2019/07/01 UTC: 10:40
--> DATE: 2019/07/01 UTC: 10:50
--> DATE: 2019/07/01 UTC: 11:00
- Found all A1      met fields for 2019/07/01 11:30
--> DATE: 2019/07/01 UTC: 11:10
--> DATE: 2019/07/01 UTC: 11:20
--> DATE: 2019/07/01 UTC: 11:30
--> DATE: 2019/07/01 UTC: 11:40
--> DATE: 2019/07/01 UTC: 11:50
--> DATE: 2019/07/01 UTC: 12:00
HEMCO: Opening /project/eesc4550/GEOS-Chem/ExtData/HEMCO/0
- Found all A1      met fields for 2019/07/01 12:30
- Found all A3cld   met fields for 2019/07/01 13:30
- Found all A3dyn   met fields for 2019/07/01 13:30
- Found all A3mstC  met fields for 2019/07/01 13:30
- Found all A3mstE  met fields for 2019/07/01 13:30
- Found all I3      met fields for 2019/07/01 15:00
--> DATE: 2019/07/01 UTC: 12:10
--> DATE: 2019/07/01 UTC: 12:20
--> DATE: 2019/07/01 UTC: 12:30
--> DATE: 2019/07/01 UTC: 12:40
--> DATE: 2019/07/01 UTC: 12:50
--> DATE: 2019/07/01 UTC: 13:00
- Found all A1      met fields for 2019/07/01 13:30
--> DATE: 2019/07/01 UTC: 13:10
--> DATE: 2019/07/01 UTC: 13:20
--> DATE: 2019/07/01 UTC: 13:30
```

```
=====
GEOS - CHEM TIMERS
=====
Timer name                                DD-hh:mm:ss.SSS    Total Seconds
-----
GEOS-Chem                                : 00-00:29:07.382   1747.383
HEMCO                                     : 00-00:02:51.445   171.445
All chemistry                             : 00-00:16:28.187   988.188
=> Gas-phase chem                         : 00-00:12:59.851   779.852
=> Photolysis                             : 00-00:01:27.429    87.430
=> Aerosol chem                           : 00-00:01:33.921    93.922
=> Linearized chem                        : 00-00:00:00.539     0.539
Transport                                 : 00-00:02:00.484   120.484
Convection                                : 00-00:01:19.187    79.188
Boundary layer mixing                     : 00-00:01:25.023    85.023
Dry deposition                            : 00-00:00:06.789     6.789
Wet deposition                            : 00-00:01:27.835    87.836
Diagnostics                              : 00-00:01:36.171    96.172
Unit conversions                          : 00-00:01:03.226    63.227

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%% To compute the OH metrics, execute the Python %%%%%%%%%
%%%%%%%% script "metrics.py" in this run directory. %%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

====> SIMULATION END TIME: 2026/01/21 19:00 <===

*****      E N D   O F   G E O S  - -  C H E M      *****
===== run ended at Wed Jan 21 19:00:28 HKT 2026 =====
```

8. Checking the Output

```
cd OutputDir
```

```
ls -lrth
```

```
[s1155189917@chpc-sandbox gc_4x5_47L_merra2_fullchem_benchmark]$ cd OutputDir
[s1155189917@chpc-sandbox OutputDir]$ ls -lrth
total 1007M
-rw-rw-r--. 1 s1155189917 eesc4550 77K Jan 21 18:32 geoschem_species_metadata.yml
-rw-rw-r--. 1 s1155189917 eesc4550 39M Jan 21 19:00 GEOSChem.StateMet.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 91M Jan 21 19:00 GEOSChem.JValues.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 5.5M Jan 21 19:00 GEOSChem.DryDep.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 100M Jan 21 19:00 GEOSChem.Aerosols.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 12M Jan 21 19:00 GEOSChem.AerosolMass.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 629K Jan 21 19:00 GEOSChem.AdvFluxVert.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 230M Jan 21 19:00 GEOSChem.SpeciesConc.20190701_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 39M Jan 21 19:00 GEOSChem.StateMet.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 91M Jan 21 19:00 GEOSChem.JValues.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 5.5M Jan 21 19:00 GEOSChem.DryDep.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 100M Jan 21 19:00 GEOSChem.Aerosols.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 12M Jan 21 19:00 GEOSChem.AerosolMass.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 629K Jan 21 19:00 GEOSChem.AdvFluxVert.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 230M Jan 21 19:00 GEOSChem.SpeciesConc.20190702_0000z.nc4
-rw-rw-r--. 1 s1155189917 eesc4550 123K Jan 21 19:00 GEOSChem.Metrics.20190701_0000z.nc4
[s1155189917@chpc-sandbox OutputDir]$
```

You may see the log files of the model to monitor the run status and to find any (if exists) errors.

After running the model, you may analyse the outputs.

A simple tool is **Panpoly** if you have downloaded your `.nc` files to your computer.